1 Introduction

GNSS is widely used for determining accurate positions on Earth. One of the fundamental techniques for GNSS positioning is absolute (or single-point) positioning, where a receiver determines its location using signals from GNSS satellites and without the need for reference stations.

Each satellite continuously transmits signals containing its ephemeris (orbital data) and precise timestamp. When a GNSS receiver captures this signal, it records the time of arrival and uses the difference between transmission and reception times to estimate the pseudorange—the apparent distance between satellite and receiver, affected by satellite clock errors, atmospheric delays, and receiver clock bias.

The satellite position at the transmission time must be known and using the broadcast ephemeris, the satellite coordinates can be computed, at the same time time that the satellite clock error is obtained, to do the necessary corrections.

With at least four satellites, the receiver can solve the system of nonlinear equations to estimate its position in cartesian coordinates X,Y,Z and its clock bias t. This is done using iterative least squares and linearization, where the observation equations are linearized around an initial guess and iteratively refined.

However, when computing the receiver's position, we should consider that the Earth is rotating and so on the satellites. The Earth's rotation means that the receiver is moving while the signal is traveling from the satellite to the receiver, this movement causes a difference in the travel time of the signal, which must be corrected to ensure accurate positioning. This effect can introduce errors in the measured pseudoranges if not corrected. The effect of the rotation is called Sagnac effect. To avoid this errors, is important to set initial coordinates in our computations, this way GNSS systems can provide precise and reliable positioning information, ensuring the accuracy needed for various applications.

The topocentric distance is the distance between receiver and satellite and it can be measure shorter because of the rotation of the coordinate system, this distances must be computed in a non rotating system.

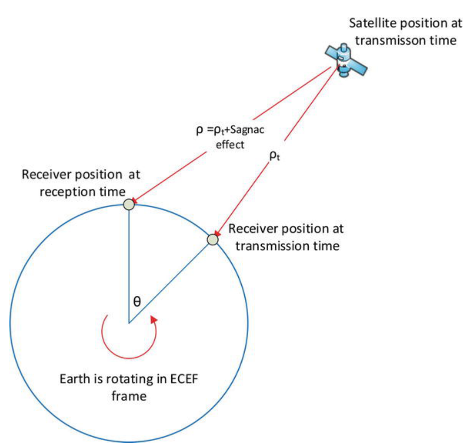

Figure 1: Effect of the rotation of Earth on the measured distances [1].

In addition to the errors introduced by the Earth's rotation, we should include the errors introduced by satellites distribution.

The geometry of the satellites relative to the GNSS receiver plays a crucial role in determining the accuracy of the computed position. GNSS positioning relies on the intersection of spherical ranges from multiple satellites. The more spread out the satellites are in the sky, the better these intersection points define a unique and accurate position on Earth.

A good satellite geometry means that the satellites are evenly distributed across the sky, meaning that they are not clustered close together in one part of the sky, so they can cover a wide range of elevation and azimuth angles. When satellites are widely spaced, the intersection of pseudorange spheres forms a well-defined position solution. This results in a low Position Dilution of Precision (PDOP) value, which indicates higher confidence in the computed position. PDOP or Position Dilution of Precision, quantifies the effect of satellite geometry on the accuracy of the position estimate.

The goal of this assignment is to determine a GPS position and the receiver clock error from code pseudoranges, in order to achieve this goal we will use the pseudoranges from different satellites, the computed satellites coordinates, satellite clocks correction from broadcast ephemeris, the tropospheric delay and ionospheric delay. All this measures where determined in previous assignments.

2 Methodology

The process of determining the receiver's position from code pseudoranges involves the use of satellites coordinates, satellites clock corrections and atmospheric corrections. The process is described in [2]. And it consists in five different steps that will be the continuation of the seven steps explained to compute the satellites coordinates and satellite clocks correction from broadcast ephemeris and the two steps that refers to the computation of the tropospheric delay and ionospheric delay.

Before starting with the computation of approximate distances, it is important to consider that the Earth and the satellites are constantly rotating (Sagnac effect), and if this is not taken into account some errors will be introduced. To eliminate this error, we must define an intermediate coordinate system called Earth Centered Inertial Coordinate System (ECI), that allowed us to define the satellites coordinates as:

$$\begin{aligned}X'^s &= X^s \cos\theta - Y^S \sin\theta,\\ Y'^s &= X^s \sin\theta + Y^S \cos\theta,\\ Z'^s &= Z^s,\\ \theta &= \dot{\Omega}_e \left(t^s - t_A\right).\end{aligned}$$

Where $\dot{\Omega}_e = 7.2921151467e - 5\text{rad/s}$ is the Earth's rotation rate, X^s, Y^S, Z^s are the satellites coordinates, t^s is the system time of the signal transmission computed from the nominal time of signal transmission measured by the clock of the satellite corrected with the computed satellite clock error, and t_A is the system time of the signal reception. The coordinates X'^s, Y'^S, Z'^s define the cartesian coordinates of the satellites in WGS84 at time t^s transformed to ECI system.

Once the satellites coordinates are set, we can start computing receiver's position,

- **Step 10. Compute approximate distances.** From the cartesian coordinates of the satellites in WGS84 at time t^s transformed to ECI system and the approximate receiver coordinates X_{A0}, Y_{A0}, Z_{A0} we compute the approximate distance between satellite and receiver as

  $$\rho^s_{A0}(\tilde{t}_A) = \sqrt{\left(X'^s - X_{A0}\right)^2 + \left(Y'^s - Y_{A0}\right)^2 + \left(Z'^s - Z_{A0}\right)^2}.$$

  Where the approximate receiver coordinates can initially be set to [0,0,0], and then by an iterative process that will be explained in the following steps, we obtain that this coordinates will converge to the correct coordinates. But to speed the process we start by choosing the approximate coordinates [3104219, 998383, 5463290], obtained from the observation files.

- **Step 11 - 13. Compute parameters by LSQ method.** The observation equation that relates the measured pseudoranges with approximate distances corrected by atmospheric and time errors, is

  $$P^s_A(\tilde{t}_A) - \rho^s_{A0}(\tilde{t}_A) + c\delta t^s_{L1} - I^s_A(\tilde{t}_A) - T^s_A(\tilde{t}_A) = a^s_X \Delta X + a^s_Y \Delta Y + a^s_Z \Delta Z + c\delta t_A.$$

  In this equation, the terms $\Delta X, \Delta Y, \Delta Z$ and δt_A are unknown, and they can be solve by the least squares method (LSQ) if at least four measurements are available.

  The matrix notation for this method is

  $$\mathbf{L} - \mathbf{v} = \mathbf{AX}.$$

In **Step 11**, we compute the vector **L**. This vector has n elements, one for each satellite, and each row contains the left side of the observation equation for each satellite where all the terms are known parameters.

$$L = \begin{bmatrix} P_A^1(\tilde{t}_A) - \rho A_0^1(\tilde{t}_A) + c\delta t_1^1 - \Gamma_A^1(\tilde{t}_A) - T_A^1(\tilde{t}_A) \\ \vdots \\ P_A^n(\tilde{t}_A) - \rho A_0^n(\tilde{t}_A) + c\delta t_n^1 - \Gamma_A^n(\tilde{t}_A) - T_A^n(\tilde{t}_A) \end{bmatrix}.$$

In **Step 12**, we compute the elements of the matrix **A**, which contains the unknown parameters a_X^s, a_Y^s, a_S^s for the n satellites. The matrix has size n x 4, and the last column is a vector of ones.

$$A = \begin{bmatrix} a_X^1 & a_Y^1 & a_Z^1 & 1 \\ a_X^2 & a_Y^2 & a_Z^2 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ a_X^n & a_Y^n & a_Z^n & 1 \end{bmatrix}.$$

The unknown terms are computed by solving

$$\frac{\partial \rho_{SA0}^s(t_A)}{\partial X_A 0} = a_x^* = \frac{-X_S - X_A 0}{\rho_{A0}^s(\tilde{t}_A)}$$

$$\frac{\partial \rho_{SA0}^s(t_A)}{\partial Y_A 0} = a_Y^* = \frac{-Y_S - Y_A 0}{\rho_{A0}^s(\tilde{t}_A)}$$

$$\frac{\partial \rho_{SA0}^s(t_A)}{\partial Z_A 0} = a_Z^* = \frac{-Z_S - Z_A 0}{\rho_{A0}^s(\tilde{t}_A)}$$

This equations are obtained from the Taylor's expansion of the linearization of the equation used in step 10 to compute the distances.

Finally, on **Step 13**, we compute we estimate the unknown parameters defined in vector **X**,

$$X = [\Delta X, \Delta Y, \Delta Z, c\delta t_A]^T.$$

The computation of the receiver's clock error term is done multiplied by the speed of light (c) to get a better numerical stability. The vector **X** is computed by solving

$$\mathbf{X} = \mathbf{Q}_X \mathbf{A}^T \mathbf{L},$$

where $\mathbf{Q}_X = (\mathbf{A}^T \mathbf{A})^{-1}$.

In this step we also compute the vector of unknown residuals **v**,

$$\mathbf{v} = \mathbf{L} - \mathbf{A}\mathbf{X}.$$

- **Step 14. Update receivers coordinates.** From the vector of unknowns, we have have obtain the value of the correction needed in the coordinates, meaning that the estimated coordinates of the receiver are finally computed by:

$$X^A = X^{A0} + \Delta X,$$
$$Y^A = Y^{A0} + \Delta Y,$$
$$Z^A = Z^{A0} + \Delta Z.$$

- **Step 15. Repetition until convergence.**

  This four steps will be repeated until $\left|\left(\mathbf{v}^{\mathrm{T}}\mathbf{v}\right)_i - \left(\mathbf{v}^{\mathrm{T}}\mathbf{v}\right)_{i-1}\right| < \epsilon$, where $\epsilon = 1e-5$ is choose in order to obtain numerical precision in the computed coordinates.

In addition, we compute the PDOP as $PDOP = \sqrt{Q_{x(1,1)} + Q_{x(2,2)} + Q_{x(3,3)}}$ and the standard uncertainties of the estimated parameters as

$$\sigma_X = \sigma_0\sqrt{Q_{x(1,1)}},$$

$$\sigma_Y = \sigma_0\sqrt{Q_{x(2,2)}},$$

$$\sigma_Z = \sigma_0\sqrt{Q_{x(3,3)}},$$

$$\sigma_{\cos t} = \sigma_0\sqrt{Q_{x(4,4)}},$$

$$\sigma_0 = \sqrt{\frac{\mathbf{v}^T\mathbf{v}}{n-k}}.$$

Where $\mathbf{Q}$ is the matrix computed in step 13 as $\mathbf{Q}_X = (\mathbf{A}^T\mathbf{A})^{-1}$.

In order to achieve our goal to compute the receiver's position from code pseudoranges, we should implement this steps in a Matlab code. To make more easy to understand and read our code, we have implement five different functions that collect all the necessary computations for steps 10 to 14 respectively. For step 15, we have create a loop that was calling this functions while doing the computations. This loop will stop when the stop condition $\left|\left(\mathbf{v}^{\mathrm{T}}\mathbf{v}\right)_i - \left(\mathbf{v}^{\mathrm{T}}\mathbf{v}\right)_{i-1}\right| < \epsilon$, is satisfied or when the maximum number of iterations reach 200.

The atmospheric corrections, satellites coordinates and satellites clock corrections where computed in previous assignments, and this time only the numerical results where taken into consideration.

To check if our results where correct, we have compare the numerical solutions without atmospheric corrections, obtained for the first iteration, against the numerical values given in the file `SPPResults.xlsx`.

3 Results

Using the approximate Cartesian coordinates [3104219, 998383, 5463290], the satellite coordinates, the atmospheric corrections, and the satellite clock correction, we computed the receiver position and the receiver clock error.

We obtained the Cartesian coordinates in meters of the receiver:

$$X^A = 3104221.323m,$$
$$Y^A = 998383.197m,$$
$$Z^A = 5463286.450m.$$

We can see a small difference of a few meters between the specified coordinates in the observation file and the computed coordinates. The biggest difference is in the Z coordinate, with a difference of less than 5 meters.

We can also calculate the receiver clock error. We multiply this value by the speed of light to better analyze it and ensure numerical stability. The receiver clock bias has a value of 155854.220 meters.

We also calculate the PDOP, or the Position Dilution of Precision, which shows how strong the satellite configuration is. In this case, we get a value of PDOP = 1.262, which is low, meaning that the positioning is more accurate.

Parameter	**Value**
Estimated X coordinate (m)	3 104 221.323
Estimated Y coordinate (m)	998 383.197
Estimated Z coordinate (m)	5 463 286.450
Receiver clock bias (m)	155 854.220
Position Dilution of Precision (PDOP)	1.262

Table 1: Estimated Receiver Position and PDOP

4 Analysis and Discussion

We have 12 satellites with known Cartesian coordinates and clock errors, and their pseudoranges are given by P1 code. We were able to compute the Cartesian coordinates of the receiver with the help of an approximate position.

The estimated receiver position and clock bias obtained from the single point positioning algorithm are:

$$\begin{aligned} X^A &= 3104221.323m, \\ Y^A &= 998383.197m, \\ Z^A &= 5463286.450m, \\ c\delta t_A &= 155854.220m. \end{aligned}$$

The RINEX observation file header typically includes the known or approximate position of the GNSS receiver. In this case, the coordinates are

$$\begin{aligned} X_{RINEX} &= 3104219.4530m, \\ Y_{RINEX} &= 998383.982m, \\ Z_{RINEX} &= 5463290.5080m. \end{aligned}$$

As we've said before, these numbers are a little different from the coordinates we calculated, especially the Z coordinate.

The absolute position error between coordinates can be computed as the difference between the RINEX coordinates and the computed coordinates, and the values obtained are

$$\begin{aligned} \Delta X &= -1.86999m, \\ \Delta Y &= 0.784999m, \\ \Delta Z &= 4.058000m. \end{aligned}$$

And the total positioning error obtained is then

$$\sqrt{\Delta X^2 + \Delta Y^2 + \Delta Z^2} = 3.686013m.$$

This error value obtained is slightly small, meaning that the algorithm computed is accurate for the computation of the receiver's position.

The Position Dilution of Precision (PDOP) value obtained was 1.262. The PDOP is an scalar value that reflects the geometric strength of satellite configuration when calculating a GPS position. It relates to how the geometry of the satellites affects the positional accuracy. For standard GPS navigation, when the PDOP is smaller than 6 the computations are accurate enough.

A value of 1.262 is small, which means that the satellites used were spread out in the sky (good geometry). This suggests that the receiver's position computed is likely very accurate, assuming that errors like ionospheric/tropospheric delays are well corrected.

The low total positioning error and the small value of the PDOP suggest that the coordinates are accurate. This is one of the key elements in GNSS systems and its applications in positioning, from everyday navigation to high-precision scientific research.

References

[1] Hasanuddin Z. Abidin et al. "DOP and Its Effect on the Accuracy of GPS Positioning". In: *Geographic Information Systems and Science*. Ed. by Fatimazahra Belouettar. IntechOpen, 2018. DOI: 10.5772/intechopen.76049. URL: https://www.intechopen.com/chapters/60049.

[2] Milan Horemuz. *GPS Single Point Positioning Algorithm*. Royal Institute of Technology, 2024.